Data collection through Webscraping

Rafiq Islam

2024-08-14

Table of contents

Introduction 1
Data Source Websites 1
Web Scraping 2
Example of Webscraping from a real website 5
References 11

Introduction

Collecting data and preparing it for a project is one of the most important tasks in any data science or machine learning project. There are many sources from where we can collect data for a project, such as

- Connecting to a SQL database server
- Data Source Websites such as Kaggle, Google Dataset Search, UCI Machine Learning Repo etc
- Web Scraping with Beautiful Soup
- Using Python API

Data Source Websites

Data source websites mainly falls into two categories such as data repositories and data science competitions. There are many such websites.

1. The UCI Machine Learning Repository

2. The Harvard Dataverse
3. The Mendeley Data Repository
4. The 538
5. The New Yourk Times

6. The International Data Analysis Olympiad
7. Kaggle Competition

Example of collecting data from UCI Machine Learning Repository

```
from ucimlrepo import fetch_ucirepo

# fetch dataset
iris = fetch_ucirepo(id=53)

# data (as pandas dataframes)
X = iris.data.features
y = iris.data.targets

# metadata
print(iris.metadata)

# variable information
print(iris.variables)
```

you may need to install the UCI Machine Learning Repository as a package using pip.

```
pip install ucimlrepo
```

Web Scraping

We scrapping is another way of collecting the data for the research if the data is not available in any repositiory. We can collect the data from a website using a library called `BeautifulSoup` if the website has permision for other people to collect data from the website.

```
import bs4                      # library for BeautifulSoup
from bs4 import BeautifulSoup   # import the BeautifulSoup object
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from seaborn import set_style
set_style("whitegrid")
```

Now let's make a html object using `BeautifulSoup`. Let's say we have a html website that looks like below

```
html_doc="""
<!DOCTYPE html>
<html lang="en">
<head>
    <title>My Dummy HTML Document</title>
</head>
<body>
    <h1>Welcome to My Dummy HTML Document</h1>
    <p>This is a paragraph in my dummy HTML document.</p>
    <a href="https://mrislambd.github.io/blog" class="blog" id="blog"> Blog </a>
    <a href="htpps://mrislambd.github.io/research" class="research" id="research"> Research <
</body>
</html>
"""
```

Now we want to grab information from the dummy html documnet above.

```
soup=BeautifulSoup(html_doc, features='html.parser')
```

Now that we have the object `soup` we can walk through each elements in this object. For example, if we want to grab the title element,

```
soup.html.head.title
```

```
<title>My Dummy HTML Document</title>
```

Since the html document has only one title, therefore, we can simply use the following command

```
soup.title
```

```
<title>My Dummy HTML Document</title>
```

or this command to get the text only

```
soup.title.text
```

```
'My Dummy HTML Document'
```

This `soup` object is like a family tree. It has parents, children, greatgrand parents etc.

```
soup.title.parent
```

```
<head>
<title>My Dummy HTML Document</title>
</head>
```

Now to grab an attribute from the `soup` object we can use

```
soup.a
```

```
<a class="blog" href="https://mrislambd.github.io/blog" id="blog"> Blog </a>
```

or any particular thing from the attribute

```
soup.a['class']
```

```
['blog']
```

We can also find multiple attribute of the same kind

```
soup.findAll('a')
```

```
/tmp/ipykernel_6928/2082462312.py:1: DeprecationWarning: Call to deprecated method findAll.
  soup.findAll('a')
```

```
[<a class="blog" href="https://mrislambd.github.io/blog" id="blog"> Blog </a>,
 <a class="research" href="htpps://mrislambd.github.io/research" id="research"> Research </a
```

Then if we want any particular object from all `a` attribute

```
soup.findAll('a')[0]['id']
```

```
/tmp/ipykernel_6928/2617565345.py:1: DeprecationWarning: Call to deprecated method findAll.
  soup.findAll('a')[0]['id']
```

```
'blog'
```

For any `p` tag

```
soup.p.text
```

```
'This is a paragraph in my dummy HTML document.'
```

Similarly, if we want to grab all the `href`s from the `a` tags

```
[h['href'] for h in soup.findAll('a')]
```

```
/tmp/ipykernel_6928/1829196164.py:1: DeprecationWarning: Call to deprecated method findAll.
  [h['href'] for h in soup.findAll('a')]
```

```
['https://mrislambd.github.io/blog', 'htpps://mrislambd.github.io/research']
```

Example of Webscraping from a real website

In this example we want to obtain some information from NVIDIA Graduate Fellowship Program. Before accessing this website we need to know if we have permision to access their data through webscraping.

```
import requests
response = requests.get(url="https://research.nvidia.com/graduate-fellowships/archive")
response.status_code
```

```
200
```

The `status_code` 200 ensures that we have enough permision to acccess their website data. However, if we obtain `status_code` of $403, 400$, or 500 then we do not permision or a `bad request`. For more about the status codes click here.

```
soup = BeautifulSoup(response.text, 'html.parser')
```

We want to make an analysis based on the institution of the past graduate fellows. Insepecting the elements in this website we see that the `div` those have `class="archive-group"` contains the information of the past graduate fellows.

```
pf = soup.find_all("div", class_="archive-group")
```

and the first element of this `pf` contains the information of the graduate fellows in the year of 2021.

```
pf[0]
```

```
<div class="archive-group">
<h4 class="archive-group__title">2023 Grad Fellows</h4>
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
<div class="views-row"><div class="views-field views-field-title"><span class="field-content"
</div>
```

Now let's make a `pandas` dataframe using the information in this page. We can make an use of the output from the above chunk. To grab the year, we see that `archive-group__title` class with a `h4` tag contains the year for all years. With `strip=True`, the text is cleaned by removing extra whitespace from the beginning and end. We need the first element so a `split()[0]` will do the job. Then we make another group called `fellows` that contains the fellows in a certian year by using the `div` and `class"views-row"`. Once the new group created, we then iterate through this group to extract their names and corresponding institutions.

```
data=[]

for group in pf:
    year = group.find(
        "h4",class_="archive-group__title"
        ).get_text(strip=True).split()[0]

    fellows = group.find_all("div", class_="views-row")
    for fellow in fellows:
        name = fellow.find(
            "div", class_="views-field-title"
            ).get_text(strip=True)
        institute = fellow.find(
            "div", class_="views-field-field-grad-fellow-institution"
            ).get_text(strip=True)

        data.append({"Name": name, "Year": year, "Institute": institute})

data=pd.DataFrame(data)
data.head()
```

	Name	Year	Institute
0	Bobbi Yogatama (Finalist)	2023	University of Wisconsin
1	Charles Gouert (Finalist)	2023	University of Delaware
2	Chen Wang (Finalist)	2023	Stanford University
3	Cheng Chi (Finalist)	2023	Columbia University
4	Eric Ryan Chan	2023	Stanford University

Now let's perform some Exploratory Data Analysis (EDA). First, we analyze the unique values and distributions.

```
# Count the number of fellows each year
year_counts = data['Year'].value_counts().sort_values(ascending=False)
# Create a DataFrame where years are columns and counts are values in the next row
year_data = {
    'Year': year_counts.index,
    'Count': year_counts.values
}
# Create the DataFrame
year_data_counts = pd.DataFrame(year_data)
```

```
# Transpose the DataFrame and reset index to get years as columns
year_data_counts = year_data_counts.set_index('Year').T

# Display the DataFrame
print(year_data_counts)
```

```
Year   2018  2023  2022  2019  2017  2015  2006  2012  2011  2016  ...  2021  \
Count    16    15    15    15    15    12    12    11    11    11  ...    10

Year   2020  2003  2014  2008  2009  2010  2005  2004  2002
Count    10    10    10    10    10     9     8     7     6

[1 rows x 22 columns]
```

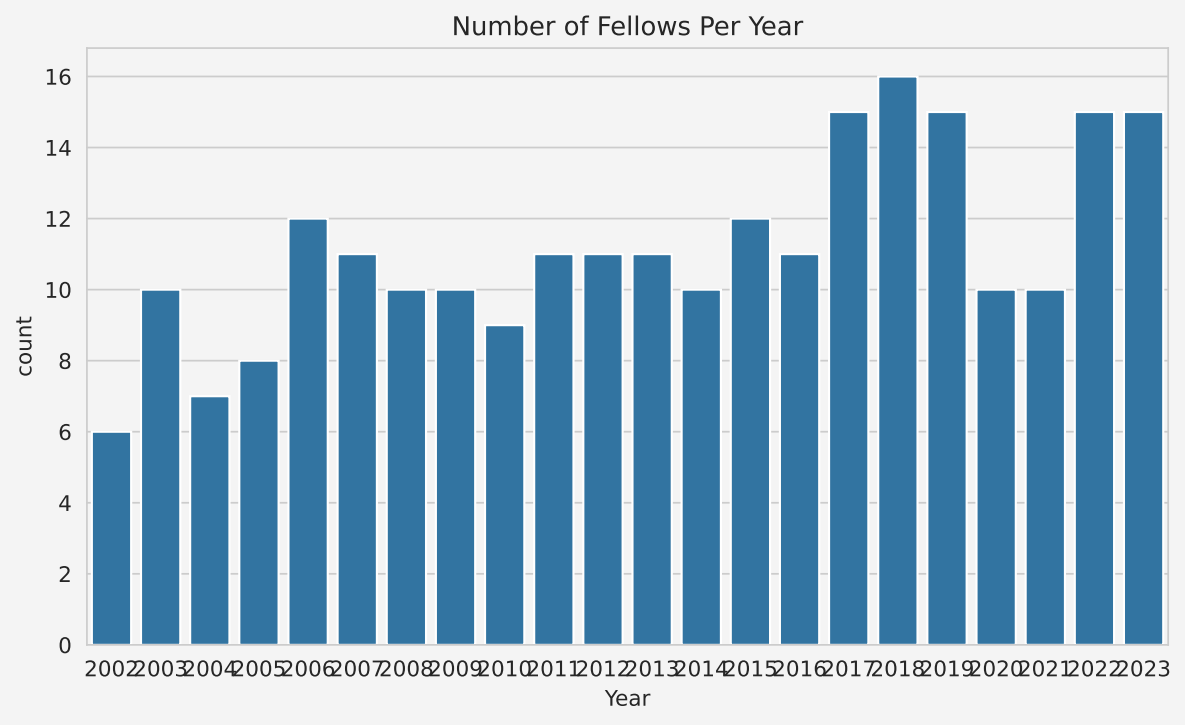
Next we see that most represented universities

```
university_counts = data['Institute'].value_counts()
print(university_counts.head(10))  # Display the top 10 universities
```

```
Institute
Stanford University                         33
Massachusetts Institute of Technology       21
Carnegie Mellon University                  18
University of California, Berkeley          17
University of Washington                    10
University of Utah                          10
University of Illinois, Urbana-Champaign     9
Georgia Institute of Technology              9
University of California, Davis              9
Cornell University                           8
Name: count, dtype: int64
```

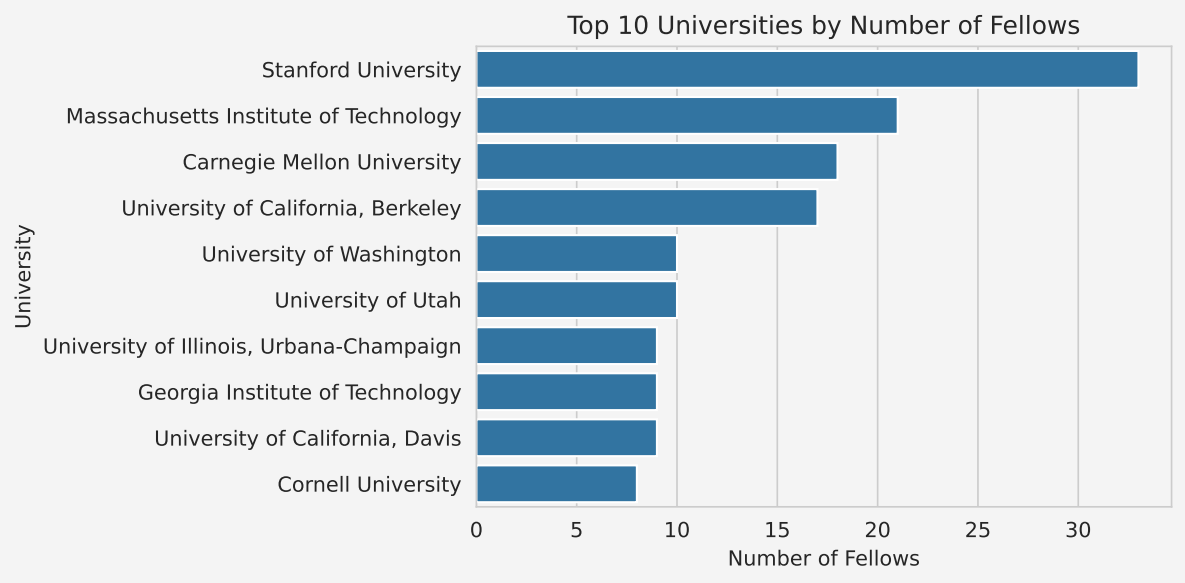
To visualize the award distributions per year,

```
plt.figure(figsize=(9,5))
sns.countplot(x='Year', data=data, order=sorted(data['Year'].unique()))
plt.gca().set_facecolor('#f4f4f4')
plt.gcf().patch.set_facecolor('#f4f4f4')
plt.title('Number of Fellows Per Year')
plt.show()
```



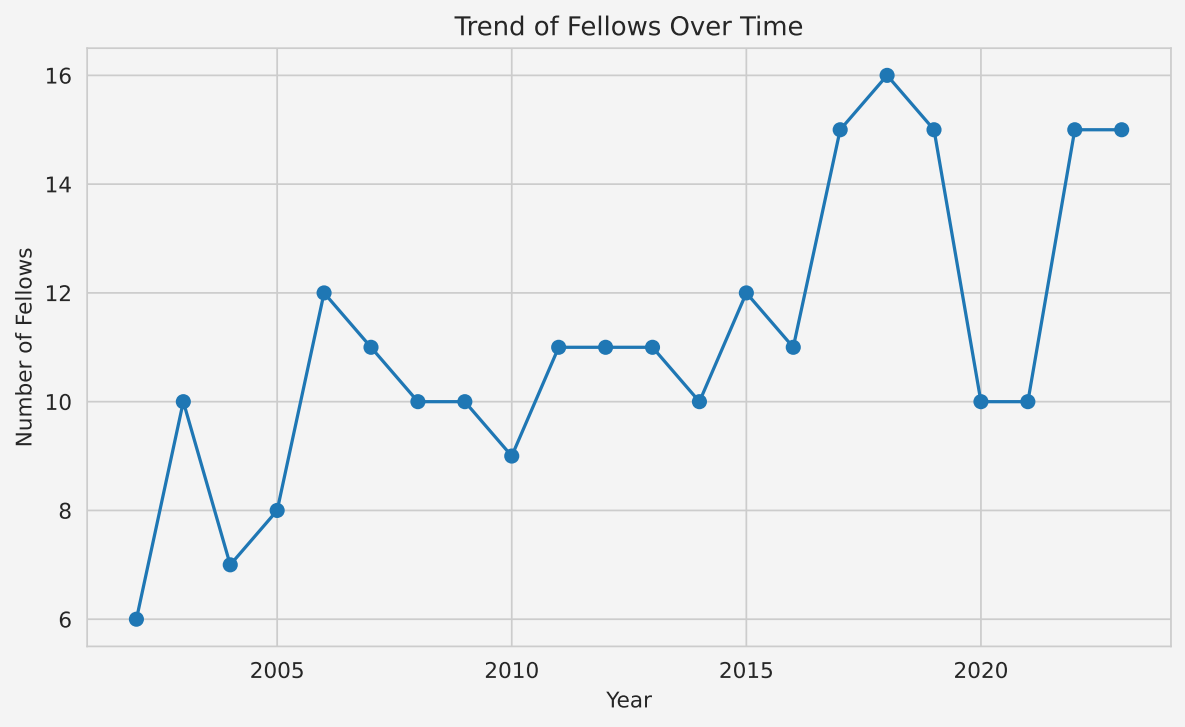
Top 10 universities visualization

```
plt.figure(figsize=(6,4))
top_universities = data['Institute'].value_counts().head(10)
sns.barplot(y=top_universities.index, x=top_universities.values)
plt.gca().set_facecolor('#f4f4f4')
plt.gcf().patch.set_facecolor('#f4f4f4')
plt.title('Top 10 Universities by Number of Fellows')
plt.xlabel('Number of Fellows')
plt.ylabel('University')
plt.show()
```



Trend over time

```
plt.figure(figsize=(9,5))
data['Year'] = data['Year'].astype(int)
yearly_trend = data.groupby('Year').size()
yearly_trend.plot(kind='line', marker='o')
plt.gca().set_facecolor('#f4f4f4')
plt.gcf().patch.set_facecolor('#f4f4f4')
plt.title('Trend of Fellows Over Time')
plt.xlabel('Year')
plt.ylabel('Number of Fellows')
plt.show()
```



This is just a simple example of collecting data through webscraping. This `BeautifulSoup` has endless potentials to use in many projects to collect the data that are not publicly available in cleaned or organized form. Thank you for reading.

References

- Fisher,R. A.. (1988). Iris. UCI Machine Learning Repository.

You may also like